

Frontier AI Companies Are Racing to Automate AI Research

Severin Field, Independent Researcher and Visiting Fellow at IAPS. Views are my own.

Key Takeaways

- This past year, I interviewed 25 research scientists at OpenAI, Anthropic, Google DeepMind, Meta, Princeton, UC Berkeley, and Stanford about AI systems being used to automate the process of AI research itself.^{1,2} I call this “recursive self-improvement” (RSI): an AI system good enough at AI development that it can design the next, more powerful version of itself, and so on.
- Making AI better at programming and AI research and then using it to build their successor has become an official plan at Anthropic and OpenAI.^{3,4,5} They say this plainly; interviewing them confirms this vision.
- 20 of 25 interviewees identified automating AI R&D among the most severe and urgent risks: AI capabilities could improve much faster than our ability to oversee, steer, or govern them. Fully automating AI R&D could set off RSI.
- The US government needs more AI capacity and visibility into AI development as we approach critical milestones.

The Most Capable Models May Never Be Public

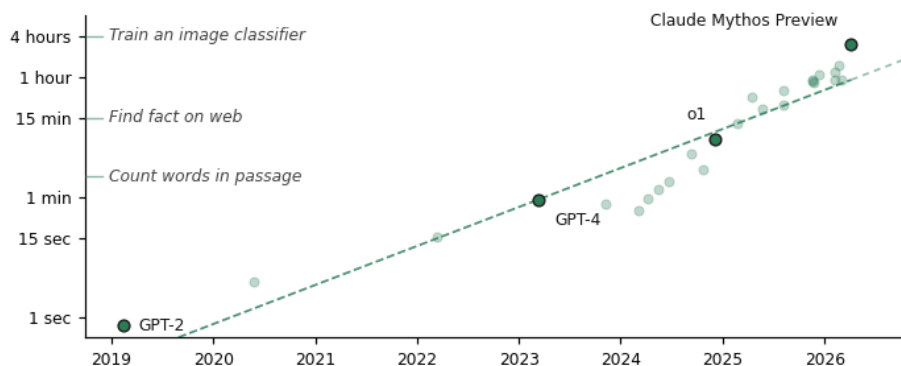
- Over half of the interviewees expected the most powerful models to increasingly be kept behind closed doors. The recent OpenAI/HuggingFace incident came from an internal-only model the public was not aware of.

The Path to Full Automation

- Recursive self-improvement is increasingly a regular company-wide discussion topic at frontier AI companies, including lead researchers and CEOs.
- Amodei: making models “good at coding and good at AI research,” using them “to produce the next generation of models... to create a loop.”⁵ Altman’s stated plan: an automated AI researcher that increasingly automates research itself.⁴
- Little consensus among interviewees on timelines or mitigations, but they generally expected a progression from coding assistants that need oversight, to systems that meaningfully contribute to research under human direction, to systems that run research and development cycles independently.
- The “Task Horizon Benchmark” maintained by METR,⁶ an independent AI evaluation organization, was consistently named the best public measure of progress. AI systems can now autonomously handle software engineering tasks that take a skilled human around 4 hours — and the time horizon doubles roughly every 7 months. If this rate holds: ~1 month of skilled software engineer labor by 2030, ~1 year a year later.

AI agents can now handle software engineering tasks that take a human hours — and that time horizon has been doubling roughly every 6.3 months.

Task length an agent completes with 80% reliability. Each dot is one AI model.



METR, Time Horizon 1.1, accessed 2026-08-03

- Half expected they already have a clear, continuous trajectory towards full automation — achievable almost entirely through scaling investment in AI hardware and algorithmic progress. Even skeptical interviewees — who believe a fundamental breakthrough is needed, such as “creativity” — agreed AI companies are racing for the “missing ingredients.”

Those Closest to RSI Are Asking for Guardrails

- In July 2026, 1,224 employees of frontier AI companies — including the chief scientists of OpenAI, Meta, and Thinking Machines, and Anthropic’s CEO and co-founders — signed an open statement⁷ warning their companies “could be close to automating AI research” and requesting the U.S. government “support an international effort to develop the . . . tools needed to deliberately pace the frontier of automated AI development.”
- OpenAI and Anthropic endorsed the statement; both have said publicly the world may need to slow down as capabilities approach critical thresholds.

Recommendations From Interviewees

- Interviewees were split on thresholds that trigger government response. One problem they flagged: the more precisely you define a threshold, the easier it is to enforce, but the worse it captures a dynamic risk that is inherently uncertain. Interviewees universally agreed on two priorities: increasing visibility and capacity.
- **Visibility:** how much the U.S. government and general public know about AI developments, especially given these might be kept internal. Often suggested: mandatory reporting requirements. Might also include:
 - Ensure government access to internal-only deployments: The America First Policy Institute warns undisclosed models available only to company staff create “a loophole in government testing and access.”⁸
 - Hold public hearings: this could include CEOs and senior researchers who see frontier AI systems months before the public. They could be asked about internal deployments and forecasts.
 - Tracking AI progress: public benchmarks or standing interview programs. Congress and the public are currently reliant on AI companies giving independent evaluators (e.g. METR) advance model access.
- **Capacity:** the government’s ability to forecast AI capabilities and build safeguards amid rapid AI progress.
 - One particularly useful capacity: the science surrounding AI treaty verification. This is because the U.S. cannot credibly propose, enter, or enforce any international agreement (e.g. nonproliferation with the PRC) without it.^{9,10}

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Notes

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4. Altman, Sam, and Jakub Pachocki. “Built to Benefit Everyone: Our Plan.” OpenAI, June 8, 2026. <https://openai.com/index/built-to-benefit-everyone-our-plan/>
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7. Pacing the Frontier, <https://www.pacingthefrontier.com/>
8. Cole Salvador and Yusuf Mahmood, “Under the Radar: How the Best AI Capabilities Could Evade Government Scrutiny,” America First Policy Institute, July 8, 2026, <https://www.americafirstpolicy.com/issues/under-the-radar-how-the-best-ai-capabilities-could-evade-government-scrutiny>.
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10. AI Futures Project, “Get Involved in Verification,” AI 2040: Plan A, updated July 9, 2026, <https://ai-2040.com/supplements/verification-plan/get-involved>